Hierarchical Scope Tables and Scope Table Merging

Compiler Sessional Online

Problem Statement

Task 1

So far we have assigned sequential IDs to scope tables. Now we have to make the IDs **hierarchical**.
Formally, if the ID of a parent scope table is `x`, then its children scopes will get IDs `x.1`, `x.2`, `x.3`, ... sequentially. The idea is visualized below -

```
1
|
|-> 1.1
|   |
|   |-> 1.1.1
|   |
|   |-> 1.1.2
|
|-> 1.2
```

Task 2

Introduce a new command `M` **(merge)**.
This is used to merge the current scope with the parent scope. Merging will move all the symbols from the current scope table to the parent scope and delete the current scope. **Merging might fail due to collision of variable names.** In that case show an error message.

Restrictions

- STL not allowed
- Must compile with `-fsanitize=address`

Examples

Sample Input

```
3
S
E
S
I a int
S
I a char
M
D a
I b float
M
P C
Q
```

Sample Output

```
 ScopeTable# 1 created
Cmd 1: S
 ScopeTable# 1.1 created
Cmd 2: E
 ScopeTable# 1.1 removed
Cmd 3: S
 ScopeTable# 1.2 created
Cmd 4: I a int
 Inserted in ScopeTable# 1.2 at position 2, 1
Cmd 5: S
 ScopeTable# 1.2.1 created
Cmd 6: I a char
 Inserted in ScopeTable# 1.2.1 at position 2, 1
Cmd 7: M
 Merge failed: 'a' already exists in parent ScopeTable
Cmd 8: D a
 Deleted 'a' from ScopeTable# 1.2.1 at position 2, 1
Cmd 9: I b float
 Inserted in ScopeTable# 1.2.1 at position 3, 1
Cmd 10: M
 ScopeTable# 1.2.1 merged with ScopeTable# 1.2
Cmd 11: P C
 ScopeTable# 1.2
 1-->
 2--> <a,int>
 3--> <b,float>
Cmd 12: Q
 ScopeTable# 1.2 removed
 ScopeTable# 1 removed
```

Marks Distribution

Task	Marks
Hierarchical ID	3
Moving variables from merged scope to parent scope	3
Deletion of merged current scope	2
Handling merge errors	2
Total	**10**